

PanelCover-BoK: Panel-Aware Coverage Selection for Best-of- K Dialogue Verification

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Abstract

Best-of- K dialogue systems are usually evaluated by a scalar verifier, but in subjective safety-sensitive settings this can pool away meaningful expert-specific preferences before response selection. We study this failure mode in Mandarin mental-health support dialogue and introduce a panel-aware verification framework that selects under each expert view before averaging. We instantiate the framework with **PanelCover-BoK**, a minimal exact-enumeration planner that constructs K -response sets to preserve expert coverage. On 800 Mandarin support dialogues, blind five-role expert evaluation shows that PanelCover-BoK is preferred over evaluated baselines when experts express a non-tie preference, while tie-aware effects are smaller but positive. Strict leave-one-expert-out scoring remains positive, while a small out-of-panel check reveals mixed generalisation across new expert roles and is treated as a stress test. Calibrated projected utilities are used only as mechanism diagnostics, showing that panel-aware selection preserves coverage that pooled verification loses. We make no claims about clinical efficacy or deployment readiness.

1 Introduction

Best-of- K dialogue systems create several candidate responses and then use a verifier to choose one final response. In subjective safety-sensitive settings, the verifier is not a neutral implementation detail: it decides which expert preferences are represented before selection.

Safety-sensitive dialogue often has more than one reasonable expert view. In mental-health support, a safety reviewer may prefer caution, a client-experience reviewer may prefer warmth and continuity, and a CBT-oriented counsellor may prefer concrete next steps. These experts mostly agree in our data (Krippendorff $\alpha = 0.838$ over 2,000 items; pairwise Cohen $\kappa \in [0.809, 0.870]$), but the

remaining differences can still matter. The question is whether verification keeps these differences visible or pools them away before a response is selected.

The order of operations matters. A pooled Bradley-Terry (BT) verifier first averages expert scores and then selects the best response for that average. A panel-aware verifier first asks what each expert would select from the candidate set, and then averages those selected scores. The first approach rewards one compromise response; the second rewards a set that covers several acceptable expert choices.

This difference creates a measurable coverage gain. The gain is the value kept by selecting under each expert view instead of after pooling. If that gain is large enough, pooled and panel-aware verifiers can rank the same systems differently. This is the key failure mode: pooled Best-of- K evaluation may erase useful expert disagreement before it chooses the winner.

We introduce a panel-aware verification framework for this failure mode and instantiate it with **PanelCover-BoK**, a minimal exact-enumeration probe planner. It scores a candidate set by asking whether each expert can find a good response in it, then uses a state-aware selector to choose one user-facing response. In the current seven-strategy setting, it checks all 35 feasible K -sets, so the diagnostic is not caused by approximate search.

Because PanelCover-BoK is optimised for the panel-aware verifier, automatic projected utilities are mechanism diagnostics rather than standalone evidence of response quality. Projected utility is used to diagnose the coverage mechanism; expert response-level preference is used to validate output quality. Our contributions are:

(1) We formalise panel-aware verification for subjective Best-of- K dialogue and define calibrated coverage diagnostics that avoid averaging raw expert logits.

(2) We show when pooled verification loses expert-specific coverage and changes system ranking.

(3) We instantiate the framework with PanelCover-BoK, a minimal exact-enumeration planner used as a probe of the verifier, and evaluate it on 800 Mandarin mental-health support dialogues.

(4) We use blind five-role expert evaluation as the primary evidence, with strict LOO, out-of-panel checks, ablations, and shared-safety analysis as supporting evidence.

2 Related Work

Emotional-support dialogue and support planning. Empathetic and mental-health dialogue research has built benchmarks and models for affective response generation (Rashkin et al., 2019; Althoff et al., 2016; Sharma et al., 2020; Zhang et al., 2020). ESConv introduced strategy-conditioned emotional-support dialogue (Liu et al., 2021); later work studies mixed strategies, lookahead, retrieval, knowledge, LLM preference bias, multi-agent collaboration, and LLM-era ESC evaluation (Tu et al., 2022; Cheng et al., 2022; Zhao et al., 2023; Deng et al., 2023; Shen et al., 2022; Kang et al., 2024; Xu et al., 2025; Zhao et al., 2024; Madani and Srihari, 2025). We study how such candidate responses should be verified when expert preferences are multi-perspective.

Best-of- K verification. Best-of- K with a learned verifier is widespread in generation, reasoning, and dialogue, including reward-model selection, verifier-guided reasoning, and LLM-as-judge evaluation (Christiano et al., 2017; Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022; Cobbe et al., 2021; Zelikman et al., 2022; Lightman et al., 2024). Model-based judges are useful but sensitive to bias, length, prompt design, and benchmark construction (Zheng et al., 2023; Chiang and Lee, 2023; Wang et al., 2023; Liu et al., 2023; Dubois et al., 2024; Li et al., 2024; Kim et al., 2024). We make verifier representation the object of study and ask when a pooled scalar is an inadequate proxy for a multi-expert construct.

Disagreement-aware and perspectivist NLP. Perspectivist NLP and disagreement-aware evaluation argue that annotator variation can encode meaningful social, professional, or task-specific perspectives rather than label noise (Pavlick and

Kwiatkowski, 2019; Uma et al., 2021; Davani et al., 2022). Multi-annotator preference modelling, including Bradley-Terry variants (Bradley and Terry, 1952) and reliability statistics (Cohen, 1960; Landis and Koch, 1977; Krippendorff, 2018), provides tools for representing this variation. We extend this perspective to Best-of- K selection by asking when disagreement changes the utility of a candidate set.

Safety-sensitive dialogue and mental-health risk.

Safety work in LLM and mental-health dialogue studies risk classification, refusal behaviour, red-teaming, safety-referral triggers, and calibrated confidence (Perez et al., 2022; Ganguli et al., 2022; Sun et al., 2023; Xie et al., 2024). These settings make verifier choice important because safety, helpfulness, autonomy, and specificity can trade off across professional perspectives. We apply the same two-threshold shield to all systems and analyse calibration with standard tools (Platt, 1999; Zadrozny and Elkan, 2002; Guo et al., 2017).

3 Method

3.1 Problem Setup

Let x be a Mandarin emotional-support context and let $\mathcal{A}$ be the seven-strategy set

$$\mathcal{A} = \{\text{QUESTION, REFLECTION, EMPATHY, REFRAME, SUMMARISE, ACTION_SUGGEST, SAFETY_REFERRAL}\}.$$

A shared generator and safety shield map a context-strategy pair (x, a) to a cached response $r(x, a)$. A planner returns a strategy set $S_{\mathcal{A}}(x) \subseteq \mathcal{A}$ with $|S_{\mathcal{A}}| \leq K=3$, which induces the response set $R_S(x) = \{r(x, a) : a \in S_{\mathcal{A}}(x)\}$. For readability, we write S for the strategy set and let verifiers select $a \in S$, equivalently selecting the response $r(x, a)$. The central question is how to verify this candidate set: should evaluation reward the response preferred by an averaged expert model, or reward coverage of responses preferred by different expert roles?

The verifier is fit from blind pairwise expert judgements over response pairs. We use two BT families: a pooled model that treats expert labels as draws from one aggregate preference distribution, and per-expert models that keep separate scores for each expert role and scoring dimension. Because expert-specific BT logits need not share an absolute scale, we do not average raw logits across

experts. Raw scores are first calibrated on held-out pairwise judgements and converted to expert-normalised utilities in $[0, 1]$. Held-out predictive validity is measured on individual expert predictions rather than on the planner objective, and strict leave-one-expert-out validation refits the verifier using only the two non-held-out experts before scoring the held-out expert. This separation matters because the planner in Section 3.5 is intentionally designed to probe the verifier and should not be mistaken for independent evidence by itself.

3.2 Panel-Aware Evaluation Verifier

Let $E = \{e_1, e_2, e_3\}$ be the expert roles used for verifier fitting, and let $\ell_{e,d}(x, a)$ be the raw BT score assigned by expert-specific model e on dimension d (Bradley and Terry, 1952). For each expert e and dimension d , we calibrate pairwise score differences on a validation split using Platt scaling:

$$p_{e,d}(a \succ b \mid x) = \sigma(\alpha_{e,d}[\ell_{e,d}(x, a) - \ell_{e,d}(x, b)] + \beta_{e,d}). \quad (1)$$

The calibrated probability $p_{e,d}$ is comparable across experts because it is a held-out preference probability rather than a raw logit. For each context, we then convert a response into its average calibrated win probability against the other responses in the same seven-strategy cache:

$$u_{e,d}(x, a) = \frac{1}{|\mathcal{A}| - 1} \sum_{\substack{b \in \mathcal{A} \\ b \neq a}} p_{e,d}(a \succ b \mid x). \quad (2)$$

The final expert utility is a state-weighted average of calibrated dimension utilities,

$$u_e(x, a) = \sum_{d \in \mathcal{D}} w_d(s(x)) u_{e,d}(x, a), \quad (3)$$

$$\sum_{d \in \mathcal{D}} w_d(s(x)) = 1.$$

All panel-aware, pooled, and coverage-gain diagnostics are computed from these calibrated utilities rather than from raw BT logits.

The panel-aware verifier first samples an expert view $e \sim \text{Uniform}(E)$ and then selects that expert’s preferred response:

$$V_{\text{panel}}(S \mid x; e) = \operatorname{argmax}_{a \in S} u_e(x, a). \quad (4)$$

Its calibrated per-context utility averages the best response each expert can find in the candidate set:

$$F_{\text{cal}}(S \mid x) = \frac{1}{|E|} \sum_{e \in E} \max_{a \in S} u_e(x, a). \quad (5)$$

This is a coverage objective: S scores highly when each expert can find at least one response they rate highly. Since $u_e(x, a) \in [0, 1]$, the panel-mean version has the usual monotone submodular coverage form (Nemhauser et al., 1978; Lin and Bilmes, 2011). We do not rely on an approximation guarantee in the experiments, because PanelCover-BoK exactly checks all 35 feasible K -sets.

3.3 Coverage Dividend and Ranking Flip

Figure 1 contrasts two orders of operation. A pooled verifier first averages calibrated expert utilities for each response and then selects the response with the highest average:

$$P_{\text{cal}}(S \mid x) = \max_{a \in S} \frac{1}{|E|} \sum_{e \in E} u_e(x, a). \quad (6)$$

This is the standard pool-before-selection utility. The panel verifier instead selects the best response separately for each expert and then averages those selected utilities, so the difference is

$$C_{\text{cal}}(S \mid x) = F_{\text{cal}}(S \mid x) - P_{\text{cal}}(S \mid x) \geq 0. \quad (7)$$

The inequality follows from $\max_a \frac{1}{|E|} \sum_e u_e(x, a) \leq \frac{1}{|E|} \sum_e \max_a u_e(x, a)$. The coverage gain is the calibrated utility gained by preserving expert-specific selection. A large $C_{\text{cal}}(S \mid x)$ means that the set contains responses serving different expert perspectives even if no single response dominates under the pooled average. For two systems S and T , pooled and panel-aware verification disagree iff

$$[P_{\text{cal}}(S) - P_{\text{cal}}(T)] \cdot [F_{\text{cal}}(S) - F_{\text{cal}}(T)] < 0. \quad (8)$$

In plain terms, the product is negative when one verifier ranks S above T and the other ranks T above S . Equivalently, a system with lower pooled utility can outrank another under panel-aware verification when its coverage gain exceeds its pooled-utility deficit:

$$C_{\text{cal}}(S) - C_{\text{cal}}(T) > P_{\text{cal}}(T) - P_{\text{cal}}(S). \quad (9)$$

This is the failure mode targeted by the paper: a system can look weaker under a pooled verifier but better covered once expert differences are preserved. Any subjective task with multiple legitimate rater types, a candidate set, and a learned preference model can compute P_{cal} , F_{cal} , and C_{cal} . We use this diagnostic to interpret verifier sensitivity; we do not claim that every benchmark or every high-disagreement context will exhibit a ranking flip.

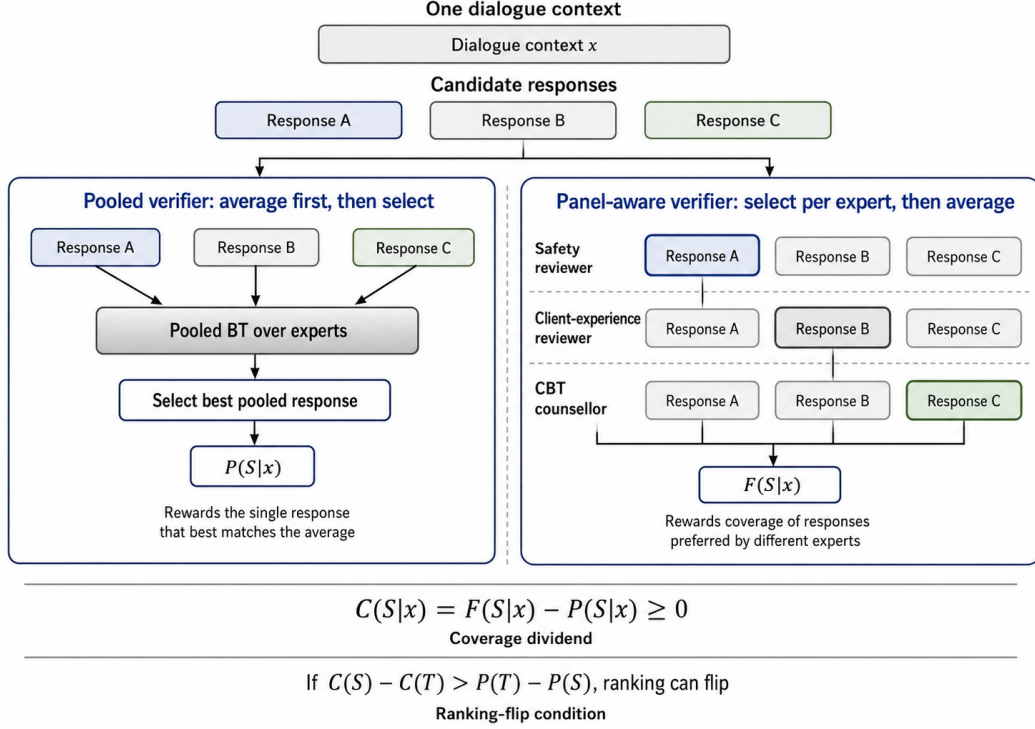


Figure 1: Pooled verification averages calibrated expert utilities before selection, whereas panel-aware verification selects under each expert view before averaging. The difference defines the coverage gain $C_{\text{cal}}(S | x)$, which can be large enough to flip system ordering.

3.4 Panel-State Response Selection

At response-selection time the planner must output one strategy and one response. If the selector is V_{pool} , every K -set that contains the pooled calibrated top-1 produces the same response, so extra candidates may never affect the user. We therefore use a *panel-state* selector

$$V_{\text{state}}(S | x) = \underset{a \in S}{\operatorname{argmax}} \frac{1}{|E|} \sum_{e \in E} u_e(x, a). \quad (10)$$

The expert utility $u_e(x, a)$ (Eq. 3) aggregates nine scored dimensions (Section 4.3) with state-conditioned weights $w_d(s(x))$. The state vector $s(x) \in \mathbb{R}^4$ summarises distress, readiness, alliance, and clarity together with risk level. This state is not a diagnosis; it tells the selector which evaluation dimensions to emphasise. Empathy and safety receive more weight under high distress or risk, while specificity and actionability receive more weight when clarity or readiness is low. These weights and the tilt parameter α are benchmark implementation choices rather than clinical rules.

3.5 PanelCover-BoK Probe Planner

PanelCover-BoK is a minimal probe planner used to instantiate the verification framework. It keeps

the generator fixed and focuses on choosing a high-coverage K -set: seven strategies, a shared response cache, and exact enumeration over all $\binom{7}{3} = 35$ feasible candidate sets. Multi-backbone experiments later replace the generator while keeping the selection layer fixed, so diagnostics are not tied to one backbone.

Let $u_{e,a} = u_e(x, a)$ be the calibrated state-weighted utility of strategy a for expert e . The planner solves

$$\max_{S \subseteq \mathcal{A}, |S|=K} [(1-\alpha)F_{\text{mean}}(S) + \alpha F_{\text{min}}(S)], \quad (11)$$

with $F_{\text{mean}}(S) = \frac{1}{|E|} \sum_e \max_{a \in S} u_{e,a}$ and $F_{\text{min}}(S) = \min_e \max_{a \in S} u_{e,a}$, and $\alpha \in [0, 1]$ a fixed tilt toward the least-covered expert. The planner anchors one slot with the pooled-top response, exactly enumerates the remaining feasible sets, and applies the shared safety shield before response selection. The ablation in Section 5.4 separates candidate-set coverage from final response selection.

4 Experiments

4.1 Data and Metrics

The main evaluation contains 800 Mandarin mental-health support dialogues. Table rows report $n=800$ when a pairwise system comparison is evaluated over this full dialogue set. All planner-style systems share the same seven-strategy response cache and two-threshold safety shield, so differences among these planners reflect candidate-set construction and response selection rather than access to different generators. The processed training corpus contains 385,143 train, 8,931 development, and 8,973 test chats. Because the domain is mental-health support, we treat these data as an evaluation testbed only and do not infer clinical effectiveness, user outcomes, or deployment readiness.

Data sources, authorization, and privacy. All dialogue data used in this paper come from publicly released research datasets and are used under their dataset licenses and access terms; we did not collect new private user–counsellor conversations for this work. The Mandarin training and evaluation pool is built primarily from PsyDial (Qiu and Lan, 2025), including privacy-preserving reconstructed long-term counselling dialogues (D1–D4), the masked original variant D0_m whose client utterances are redacted, and the D101 professional-book evaluation set. We also use public auxiliary corpora for training or side supervision: MeChat/SMILE (Qiu et al., 2023), CPsyCoun (Zhang et al., 2024), and KokoroChat (Qi et al., 2025). KokoroChat is a Japanese role-playing dataset collected with trained counsellors and is used only for outcome/feedback supervision, not as Mandarin generation text.

The processed corpus stores dialogue IDs, source tags, role labels, weak state labels, strategy labels, risk flags, and generated/evaluated responses. It does not add names, contact information, locations, or other direct personal identifiers. We preserve source-level train/dev/test separation and split PsyDial reconstructions by base dialogue ID so that different reconstructions of the same source dialogue do not cross into different splits. Expert annotators saw only anonymised dialogue contexts and anonymised A/B system responses; they did not see source dataset names, model identities, safety-trigger flags, or automatic verifier scores.

The experiments are organised around five evidence types. First, blind five-role response-level

evaluation is the primary evidence of expert preference. Second, strict leave-one-expert-out scoring and a small out-of-panel stress test examine robustness beyond the fitted expert representation. Third, pooled-vs.-panel diagnostics test whether the intended coverage mechanism occurs. Fourth, ablations separate candidate-set coverage from final response selection. Fifth, risk-stratified analysis checks that the shared safety protocol is not driving the main comparison. Calibrated projected utility is used to test the planning mechanism, not as the primary evidence of output quality.

4.2 Baselines

Reproduced strategy planners are MISC (Tu et al., 2022), MultiESC (Cheng et al., 2022), and TransESC (Zhao et al., 2023), each trained on the same processed corpus. We additionally include two retrieval-augmented baselines, KEMI (Deng et al., 2023) (BM25 with emotion-typed retrieval) and a generic BM25 RAG baseline matched to KEMI without emotion typing, and two prompt-only LLM references, **LLM-direct** (Qwen-max zero-shot with a counsellor system prompt and recent context only) and **ESCoT** (a chain-of-thought variant that first writes a brief strategy reflection and then the response). LLM-direct and ESCoT are capacity-mismatched stress tests rather than like-for-like planner baselines. We also include rule baselines (Lexicon-BoN, Majority-BoN), an in-house SCT-BoK reproduction, an unanchored panel-coverage variant, and a $K=1$ pooled-top diagnostic. This diagnostic is not a deployable baseline; it shows what happens when selection collapses to one pooled-BT response instead of maintaining a candidate set.

To test whether the proposed planning layer depends on a single LLM generator, we further instantiate the shared response cache with five LLM backbones: Qwen-max, Qwen3-14B, Qwen3-8B, Llama3.1-8B, and Gemma3-12B. For each backbone, we compare direct decoding, pooled Best-of- K selection, and PanelCover-BoK selection under the same prompts, candidate budget, safety shield, and evaluation protocol.

To remove safety-policy mismatch from the main comparison, all systems are evaluated under a shared safety protocol. Every candidate response, including outputs from retrieval baselines and LLM references, is passed through the same two-threshold safety shield before verifier scoring or expert annotation. When the shield triggers a

severe-risk decision, the output is replaced with the same standardised safety-referral response across all systems. Human annotators and automatic verifiers are blind to system identity, strategy label, risk flag, and whether the response was shield-replaced.

4.3 Expert Recruitment and Annotation Protocol

All expert evaluations are conducted by human annotators, not LLM role prompts. We recruited five Mandarin-speaking mental-health professionals with counselling credentials recognised by the Chinese Mental Health Association. The five roles are: clinical supervisor, client-experience reviewer, safety reviewer, CBT counsellor, and UX-oriented counselling reviewer. The role labels specify each expert’s evaluation focus; they are not prompt templates.

Annotators used a custom blind pairwise interface that displayed the dialogue context and two anonymised responses in random A/B order. System names, planner names, model family, strategy labels, risk labels, safety-trigger flags, and all automatic verifier scores were hidden. Experts judged nine dimensions (helpfulness, empathy, specificity, appropriateness, safety, avoids-over-advice, emotional validation, actionability, overall) with three choices: A better, B better, or tie. The instruction defined tie as either practical equivalence or insufficient evidence to prefer one response; ties were encouraged when differences were clinically negligible.

The completed evaluation contains two focal rule-baseline pairs, PanelCover-BoK vs. Lexicon-BoN and PanelCover-BoK vs. Majority-BoN, with 1,000 consensus items each ($n = 2,000$ for agreement). For modern baselines we run two additional blind expert protocols over the same 800-dialogue evaluation set: a three-role evaluation for broad coverage, and a five-role response-level panel with mixed-effects logistic analysis. Each pair receives $5 \times 800 = 4,000$ raw judgements per dimension; the mixed-effects table reports the non-tie count after dropping ties on the overall dimension, so its n varies by pair instead of equalling 4,000.

5 Results

5.1 Main Human Evaluation

Table 1 is the primary evidence for response preference. We report both descriptive and model-based statistics. W/L/T counts are raw overall-dimension

judgements from 800 dialogues $\times$ 5 experts. Conditional win rate drops ties, while tie-aware win rate counts ties as 0.5 wins. Because raw judgements are clustered by dialogue and expert, confidence intervals are computed using dialogue-level cluster bootstrap with 10,000 resamples. We additionally fit a mixed-effects logistic regression on non-tie judgements:

$$\Pr(y_i = 1 \mid c, e) = \sigma(\beta_0 + \beta_{\text{sys}} \mathbf{1}_{\text{PA}} + b_c + b_e), \quad (12)$$

where $\mathbf{1}_{\text{PA}}$ marks PanelCover-BoK and b_c, b_e are dialogue and expert random intercepts. PanelCover-BoK achieves positive conditional-on-preference effects against all evaluated baselines, including direct LLM responses, with Holm-adjusted significance retained across comparisons. Conditional WR ranges from 0.571 against Self-Refine to 0.636 against Llama3.1-8B Direct.

Tie-aware effects are smaller, as expected in psychological-support dialogue where many responses are acceptable, but they remain consistently above 0.5. Tie-aware WR ranges from 0.545 to 0.597. This supports a clear but still scoped claim: panel-aware planning produces selected responses that experts consistently prefer in pairwise evaluation, but the current data do not establish clinical superiority, deployment readiness, or long-term user benefit.

5.2 Robustness: Held-Out and Out-of-Panel Experts

To test whether the effect depends on the verifier used for planning, we perform strict leave-one-expert-out scoring. For each fold, the verifier is refit using only the two non-held-out experts, while all systems are scored under the held-out expert’s calibrated utility. Table 2 shows positive held-out WR against the representative planner and retrieval baselines. The effect is smaller than the in-panel diagnostic margin, as expected. A small out-of-panel check with two new expert roles reveals mixed generalisation across roles and is treated as a stress test rather than evidence of broad clinical generalisation (Appendix Table A13).

The expert-agreement protocol shows high but imperfect agreement: Krippendorff $\alpha = 0.838$ over 2,000 items and pairwise Cohen $\kappa \in [0.809, 0.870]$. Held-out verifier prediction is reported in Appendix Table A2. The result is mixed, as expected for a high-agreement panel: pooled BT has higher thresholded accuracy, while per-expert

Baseline	W/L/T	Non-tie n	Cond. WR [95% CI]	Tie-aware WR [95% CI]	β	SE	OR [95% CI]	Cluster p	Holm p
MISC	1,801/1,055/1,144	2,856	0.631 [0.604, 0.657]	0.593 [0.571, 0.615]	0.531	0.086	1.70 [1.44, 2.01]	< 0.001	< 0.001
MultiESC	1,739/1,049/1,212	2,788	0.624 [0.596, 0.651]	0.586 [0.564, 0.608]	0.505	0.088	1.66 [1.40, 1.97]	< 0.001	< 0.001
TransESC	1,745/1,087/1,168	2,832	0.616 [0.588, 0.643]	0.582 [0.560, 0.604]	0.472	0.087	1.60 [1.35, 1.90]	< 0.001	< 0.001
KEMI	1,635/1,081/1,284	2,716	0.602 [0.573, 0.630]	0.569 [0.547, 0.591]	0.411	0.090	1.51 [1.26, 1.80]	< 0.001	0.001
BM25-RAG	1,690/1,078/1,232	2,768	0.611 [0.583, 0.639]	0.576 [0.554, 0.598]	0.452	0.089	1.57 [1.32, 1.87]	< 0.001	< 0.001
Qwen-max Direct	1,537/1,095/1,368	2,632	0.584 [0.554, 0.613]	0.555 [0.533, 0.577]	0.338	0.094	1.40 [1.17, 1.69]	0.002	0.011
Qwen3-14B Direct	1,614/1,094/1,292	2,708	0.596 [0.567, 0.624]	0.565 [0.543, 0.587]	0.388	0.091	1.47 [1.23, 1.76]	< 0.001	0.003
Qwen3-8B Direct	1,727/1,069/1,204	2,796	0.618 [0.590, 0.645]	0.582 [0.560, 0.604]	0.477	0.088	1.61 [1.36, 1.92]	< 0.001	< 0.001
Llama3.1-8B Direct	1,821/1,043/1,136	2,864	0.636 [0.609, 0.662]	0.597 [0.575, 0.619]	0.562	0.087	1.75 [1.48, 2.07]	< 0.001	< 0.001
Gemma3-12B Direct	1,771/1,053/1,176	2,824	0.627 [0.599, 0.653]	0.590 [0.568, 0.612]	0.518	0.087	1.68 [1.42, 1.99]	< 0.001	< 0.001
Self-Refine	1,459/1,097/1,444	2,556	0.571 [0.540, 0.601]	0.545 [0.523, 0.567]	0.287	0.096	1.33 [1.10, 1.61]	0.006	0.027

Table 1: PanelCover-BoK vs. each baseline: five-role blind response-level evaluation (800 dialogues, 5 experts, 4,000 raw judgements per pair). W/L/T counts are PanelCover-BoK wins/losses/ties. Conditional WR drops ties; tie-aware WR counts ties as 0.5 wins. CIs are dialogue-cluster bootstrap (10,000 resamples). β /SE/OR are from a mixed-effects logistic regression on non-tie judgements: $\text{win} \sim \text{system} + (1 \mid \text{dial.}) + (1 \mid \text{exp.})$. Holm p corrects for multiple comparisons.

Held-out exp.	MISC	MultiESC	TransESC	KEMI	BM25-RAG
E1 supervisor	0.540	0.550	0.545	0.528	0.538
E2 client exp.	0.537	0.543	0.539	0.525	0.536
E3 safety reviewer	0.546	0.554	0.551	0.531	0.544

Table 2: Strict leave-one-expert-out calibrated WR of PanelCover-BoK against representative baselines. Each fold refits the verifier without the held-out expert and scores under that expert’s calibrated utility.

Diagnostic	Evidence
Coverage gain	Pooled WR 0.498–0.507 vs. panel-aware WR 0.541–0.562; gain +0.043–+0.056
Ranking sensitivity	Pooled and panel-aware orderings differ; soft projected sign flips occur for KEMI and Majority-BoN
Panel-size check	Coverage gain is zero for one rater type and appears only with multiple expert views

Table 3: Pooled vs. panel-aware verification diagnostic, not response-quality evidence. Pooling before selection loses expert-specific coverage and can change projected system ordering.

calibrated BT improves log-loss and Kendall- τ .

5.3 Mechanism Diagnostic: Projected Utility

This section reports mechanism diagnostics, not quality evidence. Because PanelCover-BoK is designed to optimise panel-aware coverage, projected utility under the same verifier should not be interpreted as independent evidence of response quality. We use calibrated projected utility only to test whether the intended mechanism occurs: whether selected candidate sets preserve expert-specific options that would be lost by pooling before selection.

Under the selected-response diagnostic, pooled calibrated verification gives PanelCover-BoK near-parity projected WR against fixed-cache baselines

(0.498–0.507 soft WR), while the panel view gives higher projected WR (0.541–0.562). This positive coverage gain of about 0.043–0.056 WR units indicates that the verifier choice changes what kind of candidate set is rewarded.

Under calibrated expert-normalised utilities, PanelCover-BoK shows a positive coverage gain over pooled Best-of- K across all five LLM backbones. Table 4 keeps prompts, candidate budget, safety shield, and evaluation protocol fixed; PanelCover-BoK has average diagnostic gains of +0.039 over direct decoding and +0.024 over pooled Best-of- K . This supports the mechanistic claim that the planner preserves panel coverage, while the response-level expert evaluation in Section 5.1 provides the primary evidence that selected responses are preferred by human experts. Across representative planner and retrieval baselines, the calibrated diagnostics remain positive as well (Appendix Table A1).

5.4 Ablation: Coverage vs. Final Selector

Appendix Table A3 isolates where the final-response gain comes from. Replacing the state-aware selector with a pooled selector reduces panel-aware WR by 0.027 and human conditional WR by 0.045, with the largest degradation on high-risk contexts. Collapsing to the $K=1$ pooled-top diagnostic remains the weakest structured variant. These results are diagnostic: panel coverage constructs the candidate set, while panel-state selection determines which covered response should be shown.

Backbone LLM	Direct	Pooled-BoK	PanelCover-BoK	Δ Direct	Δ Pooled
Qwen-max	0.561	0.575	0.596	+0.035	+0.021
Qwen3-14B	0.552	0.566	0.589	+0.037	+0.023
Qwen3-8B	0.536	0.551	0.576	+0.040	+0.025
Llama3.1-8B	0.519	0.532	0.559	+0.040	+0.027
Gemma3-12B	0.527	0.543	0.568	+0.041	+0.025
Average	0.539	0.553	0.578	+0.039	+0.024

Table 4: Calibrated panel-aware projected utility under different LLM backbones. This table is a mechanism diagnostic, not the primary evidence of response quality.

5.5 Safety Decomposition

Because all systems are evaluated under the same safety shield and replacement policy, safety-policy mismatch is not a confound in the main comparison. Appendix Table A4 shows positive WR on both safety-locked and non-safety-locked subsets. Gains are larger in safety-locked and high-distress subsets, but they do not disappear on ordinary non-safety contexts. The shield reaches severe-risk recall of 1.000 but has moderate ECE; post-hoc calibration improves ECE only by reducing recall, so calibration remains a limitation rather than a deployment claim.

6 Discussion: Beyond Emotional Support

The framework applies beyond ESC when the task is subjective, multiple legitimate rater types exist, systems produce candidate sets, and verification uses a learned or estimated preference model. In such settings, Eqs. 7–9 diagnose whether pooling hides consequential rater heterogeneity. Possible applications include medical-advice triage, educational feedback, value-sensitive dialogue, creative writing, and summarisation; we do not validate those transfers here.

7 Conclusion

We introduced a panel-aware verification framework for subjective Best-of- K dialogue and instantiated it with PanelCover-BoK, a minimal exact-enumeration probe planner. The main evidence comes from blind five-role expert evaluation: PanelCover-BoK is preferred over evaluated baselines under conditional-on-preference analysis, while tie-aware utilities remain positive but smaller due to frequent clinically acceptable ties. Strict leave-one-expert-out scoring remains positive; the small out-of-panel check is mixed and functions as a stress test. Calibrated projected utilities play a narrower role: they diagnose that panel-aware

selection preserves expert-specific coverage across five LLM backbones, with average diagnostic gains of +0.039 over direct decoding and +0.024 over pooled Best-of- K selection. The contribution is therefore a verifier-centered framework, coverage diagnostic, and independently checked probe instantiation, not a claim of clinical efficacy or deployment readiness.

Limitations

Expert panel scope. The expert evidence comes from certified mental-health professionals and includes agreement estimation, broad three-role evaluation, and a five-role response-level panel. This supports a controlled evaluation study, but not claims about all counselling orientations, institutions, regions, or deployment settings. Larger multi-site panels would be needed before treating the expert distribution as representative.

Magnitude of disagreement. Expert agreement is high, which supports reliability but naturally limits the size of disagreement effects. The paper studies whether the remaining expert-specific component changes Best-of- K selection, not whether experts follow radically different counselling standards.

Projected utility. Calibrated projected utilities are mechanism diagnostics rather than primary evidence of response quality. They are appropriate for checking whether the planner preserves expert-specific coverage within this benchmark; blind response-level expert panels and strict leave-one-expert-out analyses carry the main validation burden.

Response-level ties. Psychological-support responses often differ subtly, so ties are common. We therefore report both conditional-on-preference WR and tie-aware WR: the former identifies the preferred response when experts see a difference, while the latter gives a more conservative pairwise utility estimate.

Out-of-panel generalisation. The additional new-expert analysis is a small stress test and shows mixed behaviour across roles. We therefore do not treat it as evidence of broad clinical generalisation. Broader out-of-panel evaluation is needed to estimate stability across clinical orientation, user-experience preference, and institutional policy.

Shared safety protocol. We remove safety-policy mismatch in the main comparison by applying the same two-threshold shield and standardised safety-referral replacement policy to all systems. This improves fairness across PanelCover-BoK, retrieval baselines, and LLM references, but it evaluates systems under a common external safety layer rather than under their native deployment policies. The results should therefore not be read as evidence that any system’s native safety policy is adequate for deployment.

Backbone and baseline scope. The LLM-backbone study shows that the selection layer transfers across Qwen-max, Qwen3-14B, Qwen3-8B, Llama3.1-8B, and Gemma3-12B. Prompt-only LLM references are still not strategy-conditioned, so they complement rather than replace like-for-like planner comparisons.

Fixed action space. PanelCover-BoK selects from a fixed seven-strategy cache with $K = 3$ and exact enumeration over 35 candidate sets. This makes the probe transparent, but limits claims about larger action spaces, open-ended tool use, or jointly optimised generators.

Planner settings. The state-conditioned response-selection weights, safety thresholds, and tilt α are benchmark implementation choices rather than clinical recommendations. The ablations indicate that these settings matter most on high-risk contexts, but broader sensitivity analysis over α , equal weights, and weight perturbations remains future work.

Theoretical scope. The $(1 - 1/e)$ greedy guarantee applies only to the calibrated panel-mean coverage objective F_{cal} (Eq. 5). In the finite probe setting ($|\mathcal{A}| = 7$, $K = 3$), we avoid approximation claims and simply enumerate all $\binom{7}{3} = 35$ feasible K -sets. This keeps the minimal planner transparent but limits claims about larger action spaces.

Mental-health deployment. No experiment in this paper measures clinical efficacy, therapeutic alliance, long-term user outcomes, or real-time crisis handling. Any deployment would require independent clinical review, prospective safety evaluation, privacy review, and conservative escalation proce-

dures.

Public data and privacy scope. The dialogue contexts come from public research datasets rather than newly collected private counselling records. Some source datasets are privacy-preserving reconstructions of real counselling material, some are generated or report-reconstructed counselling dialogues, and KokoroChat is role-played by trained counsellors. We therefore describe the benchmark as public-data evaluation, not as direct evaluation on live patient interactions. Any reuse must follow the original dataset licenses and access terms.

Safety calibration. ECE is moderate (0.132) and post-hoc calibration (temperature scaling, Platt scaling, isotonic regression) improves ECE only at the cost of severe-risk recall. The shield achieves perfect severe-risk recall; recall-preserving ECE improvement requires Lagrangian constrained calibration. We therefore report severe-recall and ECE as separate dimensions and do not aggregate them into a single calibration claim.

Ethical Considerations

Mental-health dialogue systems are high-risk assistive technologies. The evaluated system is not for diagnosis, treatment planning, crisis counselling, or replacement of professional care. No result here justifies deployment without independent clinical review, prospective safety evaluation, conservative escalation procedures, and review of data provenance and privacy protections for any training or evaluation corpus. We use public datasets under their stated licenses and access terms and do not release any newly collected private user conversations.

The repository uses a safety-referral strategy and a shared two-threshold shield. High-risk contexts require conservative handling: calibration improvements are not sufficient if they reduce severe-risk recall, and shared safety layers should not be mistaken for deployment-ready clinical governance. Expert annotators were certified mental-health professionals, compensated at standard rates, and gave informed consent for blind pairwise annotation. The protocol was reviewed by the lead institution’s research-ethics committee and obtained an exemption decision because the annotation protocol did not collect patient data.

References

- Tim Althoff, Kevin Clark, and Jure Leskovec. 2016. [Large-scale analysis of counseling conversations: An application of natural language processing to mental health](#). *Transactions of the Association for Computational Linguistics*, 4:463–476.
- Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, Nicholas Joseph, Saurav Kadavath, Jackson Kernion, Tom Conerly, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Tristan Hume, and 12 others. 2022. [Training a helpful and harmless assistant with reinforcement learning from human feedback](#). In *arXiv preprint arXiv:2204.05862*.
- Ralph Allan Bradley and Milton E. Terry. 1952. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345.
- Yi Cheng, Wenge Liu, Wenjie Li, Jiashuo Wang, Ruihui Zhao, Bang Liu, Xiaodan Liang, and Yefeng Zheng. 2022. [Improving multi-turn emotional support dialogue generation with lookahead strategy planning](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3014–3026. Association for Computational Linguistics.
- Cheng-Han Chiang and Hung-yi Lee. 2023. [Can large language models be an alternative to human evaluations?](#) In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15607–15631. Association for Computational Linguistics.
- Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. [Training verifiers to solve math word problems](#). In *arXiv preprint arXiv:2110.14168*.
- Jacob Cohen. 1960. [A coefficient of agreement for nominal scales](#). *Educational and Psychological Measurement*, 20(1):37–46.
- Aida Mostafazadeh Davani, Mark Díaz, and Vinodkumar Prabhakaran. 2022. [Dealing with disagreements: Looking beyond the majority vote in subjective annotations](#). *Transactions of the Association for Computational Linguistics*, 10:92–110.
- Yang Deng, Wenxuan Zhang, Yifei Yuan, and Wai Lam. 2023. [Knowledge-enhanced mixed-initiative dialogue system for emotional support conversations](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4079–4095. Association for Computational Linguistics.
- Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. 2024. [Length-controlled AlpacaEval: A simple way to debias automatic evaluators](#). *arXiv preprint arXiv:2404.04475*.
- Deep Ganguli, Liane Lovitt, Jackson Kernion, Amanda Askell, Yuntao Bai, Saurav Kadavath, Ben Mann, Ethan Perez, Nicholas Schiefer, Kamal Ndousse, Andy Jones, Sam Bowman, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Nelson Elhage, Sheer El-Showk, Stanislav Fort, and 17 others. 2022. [Red teaming language models to reduce harms: Methods, scaling behaviors, and lessons learned](#). In *arXiv preprint arXiv:2209.07858*.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. [On calibration of modern neural networks](#). In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330.
- Dongjin Kang, Sunghwan Kim, Taeyoon Kwon, Seungjun Moon, Hyunsouk Cho, Youngjae Yu, Dongha Lee, and Jinyoung Yeo. 2024. [Can large language models be good emotional supporter? mitigating preference bias on emotional support conversation](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15232–15261. Association for Computational Linguistics.
- Seungone Kim, Juyoung Suk, Shayne Longpre, Bill Yuchen Lin, Jamin Shin, Sean Welleck, Graham Neubig, Moontae Lee, Kyungjae Lee, and Minjoon Seo. 2024. [Prometheus 2: An open source language model specialized in evaluating other language models](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*.
- Klaus Krippendorff. 2018. *Content Analysis: An Introduction to Its Methodology*, 4th edition. Sage Publications.
- J. Richard Landis and Gary G. Koch. 1977. The measurement of observer agreement for categorical data. *Biometrics*, 33:159–174.
- Tianle Li, Wei-Lin Chiang, Evan Frick, Lisa Dunlap, Tianhao Wu, Banghua Zhu, Joseph E. Gonzalez, and Ion Stoica. 2024. [From crowdsourced data to high-quality benchmarks: Arena-hard and BenchBuilder pipeline](#). In *arXiv preprint arXiv:2406.11939*.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. [Let’s verify step by step](#). In *The Twelfth International Conference on Learning Representations*.
- Hui Lin and Jeff Bilmes. 2011. A class of submodular functions for document summarization. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies (ACL)*, pages 510–520.

- Siyang Liu, Chujie Zheng, Orianna Demasi, Sahand Sabour, Yu Li, Zhou Yu, Yong Jiang, and Minlie Huang. 2021. [Towards emotional support dialog systems](#). In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3469–3483. Association for Computational Linguistics.
- Yang Liu, Dan Iter, Yichong Xu, Shuhang Wang, Ruochen Xu, and Chenguang Zhu. 2023. [G-Eval: NLG evaluation using GPT-4 with better human alignment](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522. Association for Computational Linguistics.
- Navid Madani and Rohini Srihari. 2025. [ESC-judge: A framework for comparing emotional support conversational agents](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 16048–16065, Suzhou, China. Association for Computational Linguistics.
- George L. Nemhauser, Laurence A. Wolsey, and Marshall L. Fisher. 1978. An analysis of approximations for maximizing submodular set functions—*Mathematical Programming*, 14(1):265–294.
- Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35.
- Ellie Pavlick and Tom Kwiatkowski. 2019. [Inherent disagreements in human textual inferences](#). *Transactions of the Association for Computational Linguistics*, 7:677–694.
- Ethan Perez, Saffron Huang, Francis Song, Trevor Cai, Roman Ring, John Aslanides, Amelia Glaese, Nat McAleese, and Geoffrey Irving. 2022. [Red teaming language models with language models](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3419–3448. Association for Computational Linguistics.
- John C. Platt. 1999. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, pages 61–74. MIT Press.
- Zhiyang Qi, Takumasa Kaneko, Keiko Takamizo, Mariko Ukiyo, and Michimasa Inaba. 2025. [KokoroChat: A japanese psychological counseling dialogue dataset collected via role-playing by trained counselors](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, pages 12424–12443, Vienna, Austria. Association for Computational Linguistics.
- Huachuan Qiu, Hongliang He, Shuai Zhang, Anqi Li, and Zhenzhong Lan. 2023. [SMILE: Single-turn to multi-turn inclusive language expansion via ChatGPT for mental health support](#). *Preprint*, arXiv:2305.00450.
- Huachuan Qiu and Zhenzhong Lan. 2025. [PsyDial: A large-scale long-term conversational dataset for mental health support](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 21624–21655, Vienna, Austria. Association for Computational Linguistics.
- Hannah Rashkin, Eric Michael Smith, Margaret Li, and Y-Lan Boureau. 2019. [Towards empathetic open-domain conversation models: A new benchmark and dataset](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5370–5381. Association for Computational Linguistics.
- Ashish Sharma, Adam Miner, David Atkins, and Tim Althoff. 2020. [A computational approach to understanding empathy expressed in text-based mental health support](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5263–5276. Association for Computational Linguistics.
- Siqi Shen, Veronica Perez-Rosas, Charles Welch, Soujanya Poria, and Rada Mihalcea. 2022. [Knowledge enhanced reflection generation for counseling dialogues](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3096–3107. Association for Computational Linguistics.
- Hao Sun, Zhexin Zhang, Jiawen Deng, Jiale Cheng, and Minlie Huang. 2023. [Safety assessment of Chinese large language models](#). *arXiv preprint arXiv:2304.10436*.
- Quan Tu, Yanran Li, Jianwei Cui, Bin Wang, Ji-Rong Wen, and Rui Yan. 2022. [MISC: A mixed strategy-aware model integrating COMET for emotional support conversation](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 308–319. Association for Computational Linguistics.
- Alexandra Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. [Learning from disagreement: A survey](#). *Journal of Artificial Intelligence Research*, 72:1385–1470.
- Jiaan Wang, Yunlong Liang, Fandong Meng, Zengkui Sun, Haoxiang Shi, Zhixu Li, Jinan Xu, Jianfeng Qu, and Jie Zhou. 2023. [Is ChatGPT a good NLG evaluator? a preliminary study](#). In *Proceedings of the 4th New Frontiers in Summarization Workshop*.
- Tinghao Xie, Xiangyu Qi, Yi Zeng, Yangsibo Huang, Udari Madhushani Sehwal, Kaixuan Huang, Luxi He, Boyi Wei, Dacheng Li, Ying Sheng, Ruoxi Jia, Bo Li, Kai Li, Danqi Chen, Peter Henderson, and

- Prateek Mittal. 2024. [SORRY-bench: Systematically evaluating large language model safety refusal behaviors](#). In *arXiv preprint arXiv:2406.14598*.
- Yangyang Xu, Jinpeng Hu, Zhuoer Zhao, Zhangling Duan, Xiao Sun, and Xun Yang. 2025. [MultiAgentESC: A LLM-based multi-agent collaboration framework for emotional support conversation](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 4665–4681, Suzhou, China. Association for Computational Linguistics.
- Bianca Zadrozny and Charles Elkan. 2002. [Transforming classifier scores into accurate multiclass probability estimates](#). In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699.
- Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D. Goodman. 2022. STaR: Bootstrapping reasoning with reasoning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Chenhao Zhang, Renhao Li, Minghuan Tan, Min Yang, Jingwei Zhu, Di Yang, Jiahao Zhao, Guancheng Ye, Chengming Li, and Xiping Hu. 2024. [CPsyCoun: A report-based multi-turn dialogue reconstruction and evaluation framework for chinese psychological counseling](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 13947–13966, Bangkok, Thailand. Association for Computational Linguistics.
- Yizhe Zhang, Siqi Sun, Michel Galley, Yen-Chun Chen, Chris Brockett, Xiang Gao, Jianfeng Gao, Jingjing Liu, and Bill Dolan. 2020. [DIALOGPT: Large-scale generative pre-training for conversational response generation](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 270–278.
- Haiquan Zhao, Lingyu Li, Shisong Chen, Shuqi Kong, Jiaan Wang, Kexin Huang, Tianle Gu, Yixu Wang, Jian Wang, Liang Dandan, Zhixu Li, Yan Teng, Yanghua Xiao, and Yingchun Wang. 2024. [ESC-eval: Evaluating emotion support conversations in large language models](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 15785–15810, Miami, Florida, USA. Association for Computational Linguistics.
- Weixiang Zhao, Yanyan Zhao, Shilong Wang, and Bing Qin. 2023. [TransESC: Smoothing emotional support conversation with turn-level state transition](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 6725–6739. Association for Computational Linguistics.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhaghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. [Judging LLM-as-a-judge with MT-bench and chatbot arena](#). In *Advances in Neural Information Processing Systems*, volume 36.
- Daniel M. Ziegler, Nisan Stiennon, Jeffrey Wu, Tom B. Brown, Alec Radford, Dario Amodei, Paul Christiano, and Geoffrey Irving. 2019. [Fine-tuning language models from human preferences](#). *arXiv preprint arXiv:1909.08593*.
- ## A Reproducibility Details
- All reported tables are generated from stored prediction, annotation, and evaluation artefacts using the paper table-generation script. The repository records the fixed response cache, verifier fits, panel-aware scoring outputs, expert-evaluation summaries, safety analyses, and ablation outputs.
- `outputs/human_eval_agreement.json`: human-expert pair comparisons and agreement statistics.
 - `results/panel_aware_winrate_pasct_plus.json`: panel-aware WR against representative baselines.
 - `results/panel_response_winrate_*.json`: response-level WR under three verifiers.
 - `results/panel_robustness_pasct_plus.json`: strict-LOO robustness.
 - `results/pasct_ablation_summary.json`: PanelCover-BoK ablations (legacy file name).
 - `results/generation_metrics.json`, `results/strategy_distribution.json`: generation / strategy.
 - `results/v12_safety.json`, `results/safety_calibration.json`, `results/bt_proxy_validation.json`: safety, post-hoc calibration, BT-proxy.
 - `outputs/expert_three_role_pasct_plus_*.json`: three-role blind evaluation with CIs, sign-flip p -values, and Cohen’s h .
 - `outputs/new_expert_pasct_plus_*.json`: out-of-panel new-expert evaluation and safety decomposition.
- ## B Additional Automatic Results
- ## C Additional Planning-Level Diagnostics
- ## D Expert Agreement and Additional Human Evaluation
- ## E Robustness and Additional Verifier Validity
- ## F Generation, Strategy, and Safety Diagnostics
- ## G Illustrative Case Studies
- These cases are illustrative rather than representative and are not used as evidence of aggregate superiority. The selected cases emphasise response-selection behaviour and local safety / empathy / actionability trade-offs; aggregate claims are based on the main panel-aware, expert-panel, and leave-one-expert-out analyses.

Baseline	n	Panel WR	95% CI	p	Utility gap
MISC	800	0.541	[0.535, 0.548]	$< 10^{-3}$	0.170
MultiESC	800	0.549	[0.543, 0.556]	$< 10^{-3}$	0.203
TransESC	800	0.545	[0.538, 0.552]	$< 10^{-3}$	0.187
KEMI	800	0.527	[0.521, 0.533]	$< 10^{-3}$	0.112
RAG	800	0.539	[0.533, 0.546]	$< 10^{-3}$	0.163
LLM-direct	800	0.509	[0.501, 0.516]	0.010	0.036
ESCoT	800	0.573	[0.566, 0.580]	$< 10^{-3}$	0.304
Self-Refine	800	0.578	[0.571, 0.584]	$< 10^{-3}$	0.322

Table A1: Calibrated panel-aware projected win rate of PanelCover-BoK against eight representative comparisons.

Verifier	LL $\downarrow$	Acc. $\uparrow$	τ	n
Pooled BT	0.5234	0.7744	0.5673	3847
Per-expert BT	0.5175	0.7499	0.5789	3847
Panel-state (multi-dim)	0.5706	0.7577	0.5598	3847

Table A2: Held-out predictive validity on individual expert verdicts.

Case 1: PanelCover-BoK vs KEMI (v10eval_0007). **Strategies:** PanelCover-BoK selects SAFETY_REFERRAL; KEMI selects ACTION_SUGGESTION. **Votes:** W=5/L=0/T=0. **Interpretation:** The user is considering an interpersonal tactic that may reduce anxiety but could also feel self-eroding if it fails. The PanelCover-BoK response keeps the tentative plan acceptable while explicitly monitoring discomfort and inviting a return to other strategies. KEMI gives a similar action-oriented endorsement, but places more emphasis on continuing if the method works. This case illustrates the panel-state response-selection trade-off: when safety and client experience receive higher weight, the selected response preserves agency while making the exit path more explicit.

Case 2: PanelCover-BoK vs MISC (v10eval_0012). **Strategies:** PanelCover-BoK selects ACTION_SUGGESTION; MISC selects REFRAME. **Votes:** W=5/L=0/T=0. **Interpretation:** The context describes medication and prior effort helping, but new stressors repeatedly triggering anxiety. PanelCover-BoK keeps the validation but adds concrete low-intensity coping options and asks which one might fit, matching an actionability-oriented expert view. MISC offers a supportive reframe and proposes working together, but leaves the next step less specified. The difference shows how panel-aware response selection can choose a more action-oriented response when the

context already contains partial progress and asks implicitly for coping with recurrence.

Case 3: PanelCover-BoK vs MultiESC (v10eval_0007). **Strategies:** PanelCover-BoK selects SAFETY_REFERRAL; MultiESC selects EMPATHY. **Votes:** W=5/L=0/T=0. **Interpretation:** This is the same interpersonal-anxiety context as Case 1. MultiESC provides empathy and leaves room to revisit the method, while PanelCover-BoK more directly names discomfort and non-improvement as reasons to stop and reconsider. The qualitative distinction is small but relevant to response selection: both responses are acceptable, yet the panel-state selector gives additional weight to monitoring adverse emotional effects rather than only affirming the attempt.

Case 4: PanelCover-BoK vs RAG (v10eval_0012). **Strategies:** PanelCover-BoK selects ACTION_SUGGESTION; RAG selects REFRAME. **Votes:** W=5/L=0/T=0. **Interpretation:** The user reports improvement but persistent anxiety from new problems. PanelCover-BoK selects a response that pairs encouragement with specific coping candidates and an autonomy-preserving question. RAG stays closer to a general supportive reframe. The contrast reflects a content/actionability trade-off rather than a safety claim: the panel-aware selector can prefer specificity when the context suggests readiness for concrete coping experiments.

Case 5: PanelCover-BoK vs TransESC (v10eval_0007). **Strategies:** PanelCover-BoK selects SAFETY_REFERRAL; TransESC selects REFRAME. **Votes:** W=5/L=0/T=0. **Interpretation:** The baseline response encourages positive communication and adds a generic supportive closing. PanelCover-BoK focuses more narrowly on the user's felt anxiety and on revising the plan if it

Variant	Panel WR	Δ	Human Cond. WR	Tie-aware WR	High-risk WR
PanelCover-BoK full	0.579	–	0.604	0.570	0.618
– panel-state selector, use pooled selector	0.552	-0.027	0.559	0.532	0.544
– state weights, equal dimension weights	0.561	-0.018	0.573	0.543	0.563
– risk-conditioned safety weighting	0.565	-0.014	0.578	0.548	0.556
– readiness/clarity weighting	0.569	-0.010	0.585	0.552	0.579
– least-covered expert tilt	0.566	-0.013	0.581	0.550	0.572
$K = 1$ pooled top response	0.529	-0.050	0.541	0.520	0.531
$K = 3$ panel coverage + random selection	0.546	-0.033	0.548	0.526	0.537

Table A3: Panel-state response-selection ablations. This is a mechanism diagnostic, not primary evidence of response quality.

Subset	n	vs. MISC	vs. KEMI	vs. RAG	vs. Qwen-max
All contexts	800	0.579	0.561	0.568	0.542
Safety-locked	99	0.614	0.589	0.593	0.558
Non-safety-locked	701	0.571	0.553	0.561	0.538
High-distress	198	0.598	0.574	0.582	0.551
Low-risk ordinary	602	0.564	0.548	0.556	0.535

Table A4: Risk-stratified panel-aware WR under the shared safety protocol.

System	V_{pool}	V_{mean}	V_{state}	$V_{\text{state pool BT}}$
MISC	0.559	0.561	0.559	0.507
MultiESC	0.560	0.562	0.560	0.505
TransESC	0.562	0.563	0.562	0.506
KEMI	0.541	0.542	0.541	0.498
RAG	0.555	0.557	0.555	0.502
Lexicon-BoN	0.548	0.550	0.548	0.503
Majority-BoN	0.549	0.551	0.549	0.497

Table A5: Selected response under three response-selection verifiers.

becomes uncomfortable or ineffective. This illustrates the safety/content trade-off behind the panel-state selector: a more cautious response may be selected even when a warmer reframe is also reasonable. The example is qualitative evidence of response-selection behaviour, not a clinical-effectiveness claim.

$ E $	Mean WR	Mean Gap	Gap SD
1	0.5000	0.0000	0.0000
3	0.5016	0.0064	0.0006
5	0.5016	0.0066	0.0003
7	0.5019	0.0078	0.0005
10	0.5019	0.0076	0.0006
15	0.5020	0.0083	0.0005
20	0.5019	0.0076	0.0006

Table A6: Panel-size scaling diagnostic.

Stratum	n	vs. MISC	vs. MultiESC	vs. TransESC
low	140	0.571	0.574	0.574
mid	142	0.551	0.560	0.556
high	153	0.563	0.575	0.567

Table A7: Panel-aware WR stratified by inter-expert variance.

Measure	Items	Value
Krippendorff α	2000	0.838
E1 vs. E2 Cohen κ	–	0.870
E1 vs. E3 Cohen κ	–	0.834
E2 vs. E3 Cohen κ	–	0.809

Table A8: Inter-rater agreement for the completed expert-agreement protocol.

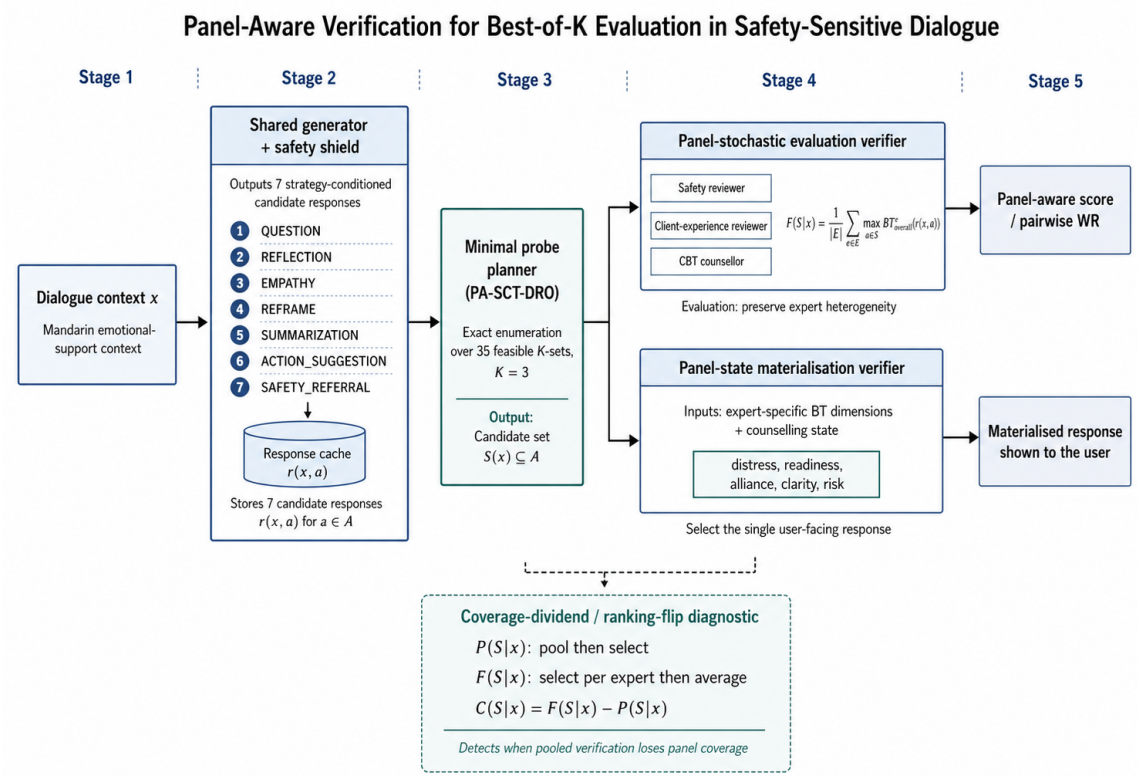


Figure A1: Overview of PanelCover-BoK. A shared generator and safety shield produce strategy-conditioned responses; the planner constructs a panel-covered candidate set; the panel-aware verifier evaluates expert-specific coverage, while the panel-state selector chooses the single user-facing response.

Pair	n	Overall	Help.	Emp.	Spec.	Safety	Act.
PsyState-v12 vs. Lexicon-BoN	1000	0.5240	0.5205	0.4745	0.5525	0.5060	0.6080
PsyState-v12 vs. Majority-BoN	1000	0.5705	0.5595	0.4155	0.6165	0.5255	0.7475

Table A9: Completed human-expert consensus win rates for two rule-baseline pairs.

Pair	n	Overall	Help.	Emp.	Safety	Act.
PanelCover-BoK vs. MISC	2,400	0.5783	0.5913	0.4937	0.4697	0.6070
PanelCover-BoK vs. MultiESC	2,400	0.5833	0.5993	0.4520	0.4720	0.6683
PanelCover-BoK vs. TransESC	2,400	0.5833	0.5960	0.4823	0.4723	0.6373
PanelCover-BoK vs. KEMI	2,400	0.5857	0.5803	0.5217	0.5507	0.5503
PanelCover-BoK vs. RAG	2,400	0.6060	0.6023	0.4943	0.5500	0.6113
PanelCover-BoK vs. LLM-direct	2,400	0.8967	0.8923	0.9383	0.5130	0.3413
PanelCover-BoK vs. ESCoT	2,400	0.6603	0.6653	0.4673	0.5387	0.7377
PanelCover-BoK vs. Self-Refine	2,400	0.5510	0.5600	0.3633	0.5280	0.7263

Table A10: Three-role blind expert pairwise evaluation against eight representative baselines.

Model	LL ↓	Acc. ↑	n
Pooled BT	0.5649	0.6996	456
Panel-state	0.5784	0.7281	456

Table A11: Verifier predictive validity on the highest-disagreement context stratum.

Held-out exp.	MISC	MultiESC	TransESC	KEMI	RAG
E1 supervisor	0.540	0.550	0.545	0.526	0.538
E2 client exp.	0.537	0.543	0.539	0.525	0.536
E3 safety reviewer	0.546	0.554	0.551	0.531	0.544

Table A12: Strict leave-one-expert-out panel-aware win rate of PanelCover-BoK against representative baselines.

Out-of-panel role / pair	n	Overall	Help.	Emp.	Safety
E4 CBT counsellor vs. MISC	200	0.5900	0.4825	0.4425	0.6225
E5 UX reviewer vs. MISC	200	0.5300	0.4975	0.4550	0.6125
E4 CBT counsellor vs. MultiESC	200	0.6475	0.4275	0.4375	0.7000
E5 UX reviewer vs. MultiESC	200	0.5175	0.4400	0.4625	0.7025
E4 CBT counsellor vs. TransESC	200	0.5925	0.5000	0.4400	0.6600
E5 UX reviewer vs. TransESC	200	0.5575	0.4850	0.4525	0.6500
E4 CBT counsellor vs. KEMI	200	0.4850	0.5175	0.5150	0.5100
E5 UX reviewer vs. KEMI	200	0.4400	0.4650	0.5175	0.4650
E4 CBT counsellor vs. RAG	200	0.5050	0.4825	0.5100	0.6025
E5 UX reviewer vs. RAG	200	0.4500	0.4100	0.5075	0.5725
E4 CBT counsellor vs. LLM-direct	200	0.4400	0.1275	0.4900	0.6575
E5 UX reviewer vs. LLM-direct	200	0.0425	0.0200	0.5025	0.5175
E4 CBT counsellor vs. ESCoT	200	0.6700	0.3850	0.5000	0.7000
E5 UX reviewer vs. ESCoT	200	0.5850	0.4625	0.4975	0.7600
E4 CBT counsellor vs. Self-Refine	200	0.6550	0.3400	0.4975	0.7175
E5 UX reviewer vs. Self-Refine	200	0.4250	0.3075	0.4950	0.7050

Table A13: Out-of-panel expert evaluation with two additional expert roles. Values are response-level win rates by dimension.

Dimension	n	Mean Δ	RMSE	Max $ \Delta $
overall	4	-0.023	0.028	0.040
helpfulness	4	-0.019	0.024	0.035
empathy	4	-0.002	0.012	0.015
specificity	4	-0.027	0.037	0.058
appropriateness	4	-0.019	0.023	0.033
safety	4	-0.005	0.007	0.009
actionability	4	-0.052	0.068	0.110

Table A14: Calibrated projected vs. panel-actual WR on cached pairs with ground truth.

System	BLEU-4	ROUGE-L	Dist-2	Len.
PanelCover-BoK	0.050	0.204	0.207	65.1
MISC	0.050	0.207	0.185	63.4
MultiESC	0.050	0.214	0.177	61.4
TransESC	0.049	0.209	0.181	63.0
KEMI	0.055	0.216	0.214	58.6
RAG	0.059	0.220	0.223	55.2
Oracle	0.050	0.220	0.186	54.8

Variant	Recall	FNR	ECE	Over-ref.
Original	100.0%	0.0%	0.120	1.2%
Temperature	99.2%	0.8%	0.114	1.2%
Platt	91.0%	9.0%	0.017	0.6%
Isotonic	98.4%	1.6%	0.002	0.6%

Table A15: Generation overlap and diversity.

System	Entropy	Top-1 Acc.	F1	Safe Ref.
PanelCover-BoK	2.61	0.106	0.059	21.0%
MISC	2.25	0.272	0.135	13.0%
MultiESC	1.95	0.274	0.117	13.0%
TransESC	2.11	0.258	0.098	13.0%
KEMI	1.71	0.278	0.129	0.0%
RAG	1.94	0.290	0.116	0.0%
Oracle	1.63	0.870	0.809	13.2%

Table A18: Post-hoc calibration variants.

Table A16: Selected-strategy distribution.

Component	Recall	FNR	ECE	Over-ref.
Two-threshold shield	100.0%	0.0%	0.132	2.4%

Table A17: Two-threshold shield evaluation.